

Cultural Game Jams with Youth: A Multiple-Site Case Study

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Figure 1: Cultural Game Jams with youth in three hubs: Left: PT1 in Óbidos. Middle: NL2 in Hilversum. Right: DK2 in Aarhus.

Abstract

In this paper, we present a multiple-site case study to illustrate the similarities and differences in cultural game jam design with youth in situated research contexts. The aim is to provide the reader with an insight into the problem space, jam design, outcomes and opportunities of a set of linked, yet individual, studies from several different contexts and countries. The cases provide a platform to think about how game jam studies might be collectively reported and evaluated even when they are carried out in different ways. The contribution is an illustration of differentiated replication in multiple cultural game jam studies with youth in different countries. The reflections on replication and differentiation is of interest for cultural game jam activities to empower youth's cultural participation through technology design.

CCS Concepts

• **Human-centered computing** → **Interaction design theory, concepts and paradigms.**

Keywords

Game jam, culture, values, design, digital games

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1 Introduction

We are witnessing how cultural diversity and inclusivity within the global game industry are still lagging behind its ever-diversifying reality [31]. One way to tackle this is to investigate the potential interrelationships or collaborations between games based on regional cultural heritage(s) and the GLAM (Galleries, Libraries, Archives, Museums) sector and through the "creative toying" with cultural heritage aspects in game development [40]. However, the definition

of what precisely constitutes cultural heritage is diverse [40]. Cultural heritage can take many forms, both tangible and intangible, and the boundaries between cultural and natural heritage are at times fuzzy. However, tangible cultural heritage can, for instance, be historical sites, artworks, monuments, digital archives, libraries, or other spaces and artefacts, but also the natural environment, which includes landscapes, flora and fauna, and geological elements [24]. There is, furthermore, a diversity of factors that deeply characterize our cultural heritage and have a non-physical nature, namely social values and traditions, customs and practices, philosophical orientations and religious beliefs, artistic expression, language and folklore [24]. Moreover, games today are by some academics considered part of cultural heritage itself (e.g., [4])

The practice of game development requires highly creative and specialised skills ([3, 18]), which are mixed with unique local culture, context, and social norms [19, 21, 40] alongside each game developer's design values and motivations [19]. Game jams are important tools for learning various skills needed for game development [32], and is an emerging widespread phenomenon for exploring multiple themes through game development [8, 32]. This includes game jams aiming to learn about cultural heritage through serious games [2, 9, 24], widening perspectives through indigenous (e.g. Sami) worldviews in digital games [26], and support empathy and compassion in Nigeria [35], to mention a few.

This paper is based on a European project¹ where cultural heritage institutions, small game enterprises from the creative industries, and universities in three countries (Denmark, The Netherlands, Portugal) collaborate with youth in cultural quadruple helix innovation ecosystems [1, 7, 28, 36] to develop, test, and evaluate a series of 12 design interventions of 'Cultural Game Jams' (CGJ) for youth. The objective of the project is to investigate the CGJ format as a possible pathway to foster youths' empowered participation as audience and co-creators in culture, cultural heritage and cultural heritage institutions - domains in which young people do not necessarily see themselves as belonging or engaged in [38].

The project is divided into two cycles, where the first six CGJs have a more local focus, which will then feed into the next cycle of six CGJs of more global character. Initially, a shared CGJ toolkit [27, 29], practical protocol [17] and research protocol [37] were defined, including ethical procedures and an observation grid. However, the project partners recognized the potential challenges in organizing and coordinating 12 interconnected game jams, where each experience builds on the previous one with the aim of fostering mutual learning and exchange. Unanswered questions remained about local and global value, what participants would learn, and how successfully the local activities would collectively meet the project's main goals, including youths' empowerment, participation, and democratization in culture.

This case study paper explores the practicalities of designing, experimenting with, and making sense of the first six locally arranged game jams, along with a pilot study [11], for the continued process toward designing and executing CGJs of a more global character.

2 Method

The project applies design-based research [23] to develop the format and materials for CGJ, entailing an iterative process that continuously materialises, tests, evaluates, and revises it through insights from the design interventions. The iterative development of the CGJ structure, materials and toolkit has been inspired by various sources, e.g. [14, 34] and value-sensitive design (VSD) [13]. Toolkits in human-computer interaction (HCI) examine new conceptual, design or technical solutions to unsolved problems. They are generative platforms designed to create new interactive artifacts, provide easy access to complex problems, enable fast prototyping, and enable creative exploration of design spaces [15, 20]. The series of design interventions share the same design brief and CGJ toolkit [29]; however, they are executed with different people in three different countries and with local adaptations to the whole CGJ facilitation. Here, we describe the interventions as a multiple-case study [39]. A case study is an empirical inquiry investigating a contemporary phenomenon [39]. In a multiple-case study, researchers study multiple cases to understand similarities and differences [39]. Each case must be carefully selected to (a) predict similar results or (b) predict contrasting results but for reasons that can be anticipated [39]. In contrast, *differentiated replication* involves deliberate, or at least known, variations in fairly significant aspects of the study conditions. It is a search for exceptions, a route toward establishing the conditions under which the generalization does not hold [22]. The various CGJs share the same toolkit and research protocol, where the data have been collected through pre- and post-questionnaires, observations, interviews, and game documentation (devlogs) in itch.io. The analyses supporting this paper were developed and synthesised through initial local and later cross-cutting analyses, drawing on thematic analysis [6], to synthesise findings across sites while enabling the researchers' reflexivity and contextual knowledge and reducing bias. For an overview of the six CGJs, see Table 1. In addition, all CGJs share ethical practices, where ethical consent for the study was obtained from the European Union and each university's ethics board. Participants signed informed consent forms prior to the CGJ.

To analyse across a multiple-site case study, we have used the *framework for game jams in applied contexts* [33], a holistic framework for coordinating and assessing applied game jams. The framework considers four dimensions in facilitating and evaluating an effective applied game jam: 1. Identifying an issue or outcome to be considered ('Problem Space'); 2. Defining an appropriate game jam format ('Jam Design'); 3. Establishing, maintaining, and collating community engagement with the game jam ('Jam Delivery and Outcomes'); 4. Assessing the game jam against predefined outcomes and identifying additional partnership opportunities ('Follow-on Opportunity'). Although we have not seen the *framework for game jams in applied contexts* utilised in multiple-case studies before, we hope that the example presented in this paper can inspire others.

3 Results

The Applied Jam Framework analysis results are presented according to four dimensions in facilitating and evaluating an effective applied game jam [33].

¹EPIC-WE <https://epic-we.eu/>

Code	Context	Participants	Time	Theme	Material	Games
DK1	Art museum	23 youth (ages 15-27)	30h Two days weekend	Art: Human Nature	Unity, Godot CGJ design kit	5
PT1	Bookstores & Art	26 youth (age 18-28)	3+3+32h Five days weekdays	Óbidos Town of Literature and European Values	Unity, Unreal CGJ web-resources	8
NL1	Media museum	15 youth (ages 20-28)	30h Two days weekend	Open	Paper Prototyping CGJ design kit	4
PT2	Natural Lagoon	40 youth (age 15-25)	8+32h Five days weekdays	Lagoon as Cultural Heritage and European Values	Unity, Unreal CGJ design kit	14
DK2	Art museum	39 youth (age 14-25)	20h Two days weekend	Art: 1960-now	Unity, Godot CGJ design kit	9
NL2	Media museum	61 youth (ages 15-16)	20h four days weekdays	Media and Democracy Equality	Paper Prototyping CGJ design kit	6

Table 1: Overview of the six cases of Cultural Game Jams in Denmark (DK1 & DK2), Portugal (PT1 & PT2), and The Netherlands (NL1 & NL2). The cases are listed in chronological order, and took place from February–September 2024

3.1 Problem Space

The CGJs' shared brief is to design a value-based cultural game prototype and/or concept based on a *value* and *cultural heritage* related to a *societal challenge*. The series of CGJs share the same intangible culture—namely, the European *values* [12] (democracy, equality, the rule of law, human dignity, human rights, freedom). In addition, each CGJ has a specific, more locally situated *cultural heritage* theme. In **DK1**, the theme was Human Nature, which is in line with the exhibition at the art museum where the game jam was located, while in **DK2**, the theme was Art from 1960 to now. In **NL1**, the theme was open, while in **NL2**, the theme was Media, democracy and equality. In **PT1**, the theme was Literature and European values, while in **PT2**, the theme was Lagoon as cultural heritage and Sustainable Development Goals. The related *societal challenge* chosen by the youth is typically something relevant to them, such as social, environmental, economic, or cultural issues.

3.2 Jam Design

The Cultural Game Jam Process Model (see Figure 2) is developed as a progression and taxonomic model that scaffolds the CGJ process for facilitators and participants [29]. It comprises four distinct CGJ phases: *Experience*, *Play*, *Imagine* and *Create*. These phases follow a structured progression where each phase builds upon the previous, combining divergent and convergent thinking to ideate and design games, game ideas, and concepts using cultural heritage and values as game jam material.

In the CGJ process, Transition Tools play a vital role in mediating and supporting the transition from one phase to another, using physical objects and tangible processes to facilitate it. The *Ideation Wheel* (see right in Figure 2) is a hands-on, exploratory tool that bridges the first and second phases of the CGJ. Drawing on similar design tools from play research and educational research and development helps to support the ideation process to include values, culture, and citizenship into the game concept and to position the design choices. The *Expert Council* is designed to support the

transition from the second to the third phase of the CGJ, enable feedback for improvement, enhance team culture, and foster a positive and productive atmosphere. It ultimately contributes to the refinement of design ideas. The feedback received during critiques encourages designers to think creatively and may lead to more innovative choices in the design process. The *Playtest* supports the transition from the third to the last phase of the CGJ and is crucial for achieving a balanced and enjoyable gaming experience. It facilitates the collection of valuable information through feedback forms, enabling designers to refine and enhance the gaming experience. The *Log & Pitch tool* is the final transitional tool in the CGJ process model. It moves CGJ participants from the game design process into the final Exposition. This involves creating a game login and a design document to ensure effective communication and pitching of the overall games through and for culture among team members, which is shared in itch.io. The instructions for game logs were provided so that the teams could reflect on their design decisions and meet the research project's objectives, which, among other things, aims to build an online library of documented values-based cultural games. Overall, these transition tools are described in the CGJ Playbook and aim to help participants move forward through the different phases of the CGJ with greater focus, clarity, and purpose.

In addition, the participants were provided with inspirational design methods in each phase. For instance, in *Explore*, the design methods cards included values storming (a variation of brainstorming), culture safari, and value change. In the play phase, examples of methods were value collage, 6-8-5 game sketching, and storyboarding. For *Imagine*, cultural prototyping (emphasising provocative prototyping) and rapid mockup were provided. Finally, for *Create*, various forms of prototyping methods were provided.

The CGJ **DK1** ran for two days during a weekend in the AROS art museum (see Figure 1). It was organised in collaboration with partners from the university, the creative industry, and the AROS art museum. The roles and responsibilities were shared and coordinated among the various project partners, while one researcher

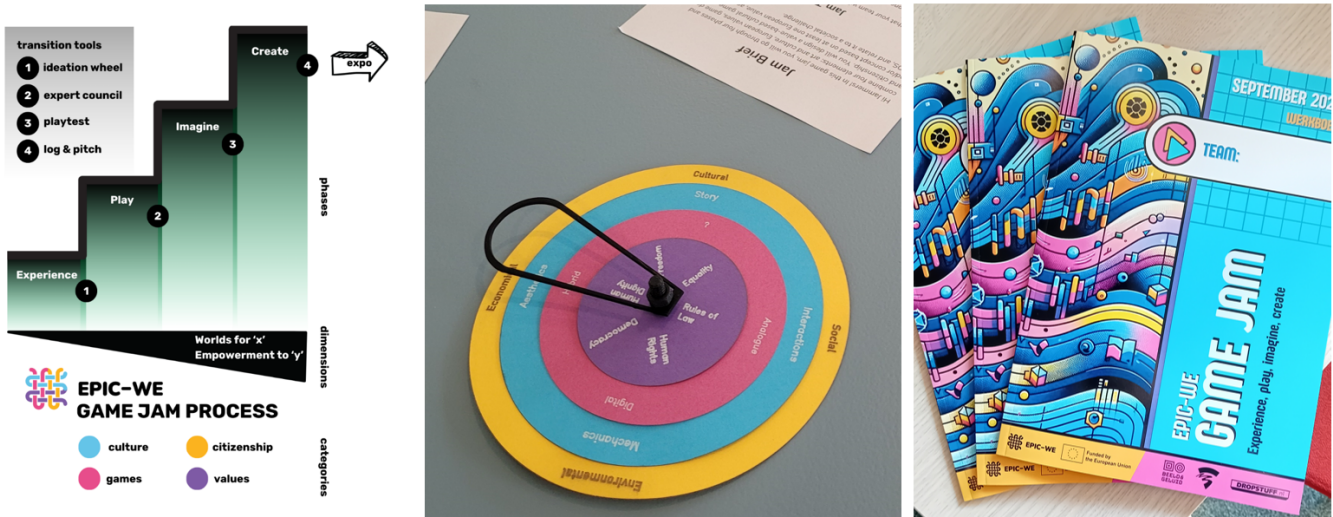


Figure 2: Left: The general Cultural Game Jam Process Model. Middle: Ideation Wheel in DK2. Right: CGJ Playbook in NL2.

was the main facilitator. The game jam program included a guided art walk in the art museum’s gallery, an exhibition entitled Human Nature, and inspirational talks by a game industry representative, an experienced game jam organiser, and a professional programmer. **DK2** ran in the same context and in a similar way as **DK1**; however, it had some changes. The artwork was more modern, the time format was shorter (from 30 to 20 hours), the information and facilitation were reduced and complemented with more group supervision, and one youth was now part of the facilitation team.

The **NL1** CGJ ran for two days during a weekend at the Netherlands Institute for Sound & Vision, a media museum with an audio-visual archive, with participants aged 20-24. The CGJ was organised by the media museum with partners from the creative industries and university researchers. The roles and responsibilities were shared and coordinated between the various project partners. The university partners focused on collecting research data through observations and video interviews and facilitating the game documentation (on itch.io) by the participants. The creative industries partner and the media museum jointly facilitated the design process of the game jam, which kicked off with a guided tour of the museum. Participants were invited to explore the museum and find inspiration for their game (either on a thematic level or for the game dynamics). Within their teams, they were allowed to choose a theme they found interesting to work on. During the CGJ, the participants worked from an initial concept to a playable (paper or digital) prototype, which was improved after rounds of expert feedback and peer play testing. At the end of the CGJ, the participants presented their final prototypes to a jury of experts from creative industries.

Because of a much younger age group (15-16), the **NL2** ran in a different format from **NL1** and was hosted primarily at the participants’ high school. One 2-hour kick-off session and three full weekdays were allocated for the CGJ during school days, totalling 20 hours. One day occurred at the media museum, while the other three days were hosted at school. The themes the participants

worked with were media, democracy, and equality. The design process was simplified compared to **NL1** by reducing the number of ice-breaking and brainstorming exercises and streamlining the documentation requirements.

In **PT1**, the theme was Literature and European Values. The jam lasted four full days (at least 32 hours), and participants had the freedom to manage their time. Pre-jam activities included the Cultural Heritage Institution presenting the meaning of “town of literature” to the recruited participants and an ideation exercise conducted by the researchers. During the game jam, game designers and developers from the university and the creative industry partner acted as facilitators. Four researchers made general observations of all participant groups from a distance to avoid interrupting the creative flow. The cultural facilitators were not present during game-making. In **PT2**, the theme was a lagoon town, a cultural heritage, and Sustainable Development Goals (SDGs). The CGJ ran for four weekdays, lasting at least 32 hours. Pre-jam activities with a duration of 8 hours were conducted jointly by the organisers, one month before the CGJ, namely a presentation of the SDGs at the researchers’ facilities and a visit to the lagoon organised by the cultural heritage institution. Cultural facilitators and experts in the lagoon fauna and flora and the lagoon history explained how the lagoon is essential for the region’s sustainability. While the format of **PT1** and **PT2** followed the same protocol, the presence of cultural facilitators during the game jam was new. Two researchers administered the participants surveys, conducted youth interviews and made observations.

3.3 Jam Delivery and Outcomes

The CGJ organisers in the project make up a new cultural model, a cultural quadruple helix innovation ecosystem [1, 7, 36], to co-create for cultural value and societal change together with youth. This helix system implies that university researchers, creative industries and cultural heritage institutions enter into a co-creative system and embrace their role of engaging youth as part of civic society

in democratic, participatory processes. The project's proposition for quadruple helix cultural innovation ecosystems of transdisciplinary collaborations opens up opportunities for new value chains to emerge. It supports new paths for knowledge creation between civil society, academia, cultural institutions, and the creative sector. Through its activities, the project aims to ensure that this potential is fully exploited and will lead to long-term impacts that benefit all target groups.

In **DK1**, the participants were recruited from various student and interest-based networks related to the organisers. There were 23 participants within the age range 15-27 (10 men, 7 women, 6 other), and they came from a background in, e.g. audio design, visual design, programming, escape room design, animation, and dramaturgy. Due to various nationalities, the primary language of the CGJ was English. Prior to the CGJ, the organizers divided the participants into five design teams based on age, gender, background, interests, and skills. Five digital games were produced and exhibited by the end of the CGJ and documented at itch.io². All five design teams designed games in which tangible and intangible cultural heritage was used as game-making material in quite different and innovative ways. In **DK2**, the recruitment followed the same line; however, one youth was quite active in the recruitment process this time. This led to 26 participants, similar to DK1, and 13 from an upper secondary high school class. Six digital games and three analogue games were produced and exhibited by the end of the CGJ and documented at itch.io³.

In **NL1**, the participants were recruited via direct communications through the networks of the organisers and social media. There were 15 participants aged 20-28 (6 men, 8 women, 1 other) with backgrounds in Software Development, Communication & Design, Media Studies, Game Design, and other disciplines. Because of the various nationalities of both the participants and the team of facilitators, the primary language of the CGJ was English. During registration, the participants were able to indicate a theme they would like to work on during the CGJ, such as climate change or mental health. At the start of the CGJ, multiple ice-breaker activities supported group formation based on the participants' mutual interest in a theme. In total, three board games and one digital game were produced and documented in itch.io⁴. The created games let players interact and learn in playful ways about societal themes important to the participants, such as climate change and mental health.

For **NL2**, the aim was to work with a lower age group (around 16 years old). For this reason, the recruitment happened differently. Instead of an open recruitment process similar to NL1, a collaboration was set up with a high school with a (graphic) design and digital media profile. This collaboration resulted in 3 groups with 61 participants in total, within the age range of 15-16. Participants of NL1 were invited to join the expert jury for the final presentations of six games documented in itch.io⁵.

In **PT1**, the participants were recruited from the university partner. The youth formed their own 8 teams, while 3 participants specialised in sound and programming and acted as "freelancers,"

meaning they were available to collaborate with all the teams that needed their expertise. A total of 26 youth (age 18-28) participated in the CGJ (5 women, 21 male). Although they were from different nationalities, the language in the CGJ was Portuguese. Their background was in science and technology, visual arts and digital games. A total of eight games were submitted by youth to itch.io⁶ and presented to organisers at the end of the CGJ.

In **PT2**, 40 participants took part, including 25 students from the university and 15 from the secondary school Josefa de Óbidos. Participants' age varies between 15-25 years (31 male, 6 female, 2 other). Their background was in science and technology, visual arts, video games, languages and humanities, and professional education. A total of fourteen digital games were submitted to itch.io⁷.

3.4 Follow-on Opportunity

One significant suggestion for change from **DK1** to the next is establishing closer collaboration with youth. According to the quadruple helix innovation system [1, 7, 36], we have so far mainly focused on collaboration between cultural heritage institutions, creative industries and universities in designing CGJ for youth. In the next step, we focus on incorporating the youth into the collaboration through a participatory design approach [5], where representatives from the youth will co-design CGJs. Further changes include reduced paperwork, balancing the documentation of the game development (devlogs), having the Expert Council earlier in the first day to support iterations better, having the youth choose an artwork before the curators provide more information about it, and including values and citizenship in practical facilitation [10].

To follow up on these suggested changes from DK1, in **DK2** a representative for youth (25 years) was recruited from the participants in DK1, which meant that all further decisions in designing DK2 were discussed with representatives for the whole quadruple helix innovation system. This particular youth helped reduce the amount of information and paperwork, balanced the research needs with practical needs, took part in recruitment and acted as an important part of the facilitator team during the CGJ. Since this youth happen to identify as queer, this led to the recruited participants being a somewhat mixed group with several queers, people we typically do not see in more traditional game jams. It also led to a higher degree of inclusion, diversity and accessibility in the actual CGJ facilitation. Examples include using pronouns on name tags, being explicit about available silent rooms for breaks, not working during the night, etc. However, because the participants were both from a formal educational setting (n=13) while others were there in their spare time (n=26), we can see a difference in motivation between the two groups. The participants who signed up voluntarily were much more motivated, while the high-school students had less motivation, which caused some challenges for the facilitation. Also, for the CGJ, there are further development opportunities for iteration when it comes to facilitating CGJ in formal educational settings, especially for students in education with no particular technical background. The students ended up designing analogue games since they were not encouraged in the

²DK1 <https://itch.io/jam/epic-we-game-jam-01-aarhus/>

³DK2 <https://itch.io/jam/epic-we-02-aarhus>

⁴NL1 <https://itch.io/jam/epic-we-01-hilversum>

⁵NL1 <https://itch.io/jam/epic-we-02-hilversum>

⁶PT1 <https://itch.io/jam/epic-we-01-obidos/entries>

⁷PT2 <https://itch.io/jam/epic-we-02-obidos/entries>

CGJ format to try something else, and they were not mixed with the other groups with more technical competencies. This is a lost opportunity to empower them in creatively jamming with culture, values, and digital technologies to become makers and co-creators of culture.

One of the main lessons learned from **NL1** was to reduce and simplify the activities during the CGJ. This includes both the game design process activities and the documentation. The participants mentioned they felt there was not enough time to work on the prototype and that they were unable to finish all the documentation during the CGJ. Many teams had to finish the documentation in the days after the CGJ. Similar to **DK1**, the **NL1** mainly collaborated with cultural heritage institutions, creative industries, and university researchers. To better include youth in the design process for the CGJ, the participants were first asked to join a Youth Advisory Board. Secondly, the participants were invited to a focus group to reflect on their experience and were asked to give feedback on the plans for the second (**NL2**) CGJ. Finally, the participants from **NL1** were invited to join the **NL2** CGJ as experts and provide feedback on the newly developed games. In **NL2**, there was a clear difference in the participants' motivation. This CGJ was a collaboration with a high school, and the pupils needed to participate as part of a school assignment. A complicating factor was that the participants were not adequately informed about the activity in advance, which made it challenging to motivate them, particularly during the first half of the CGJ when it was unclear what to expect throughout the week. Another complicating factor was in the hosting of the event. While the CGJ was planned to be hosted mainly at the media museum, due to unforeseen circumstances, the students were not able to travel, so the jam was moved to their school instead; more time spent at the museum might have led to a stronger connection between the games and the museum collection. Participants expressed excitement about the day at the media museum in the interviews. Ultimately, all participant teams created a board or digital game prototype and presented it to a jury. While the themes of media, democracy, and equality are present in the games the participants made, there are no direct references to the museum collection.

The youth participating in **PT1** were positive about the pre-jam activities and the objectives they achieved during the CGJ. A cultural visit to Óbidos (town of literature) on the 1st day of CGJ, supported by the cultural institution partner, as well as the freedom participants have to go to their specific point of interest in Óbidos exploring the theme, was considered by youth as a very positive outcome. In terms of logistics, being at Óbidos for the CGJ was appreciated. As one participant expressed: *"I am working on the game's art, and being here in Óbidos and seeing everything inspires not only the overall design but also the interfaces, characters, and style."* From the organiser's reflection and a brief analysis of the games made, it was decided that more cultural facilitators are needed during the next CGJ. Another conclusion is to consider more diversity in recruitment. The most significant outcome from **PT2** was the reinforcement of cultural facilitators throughout the event - from guiding cultural visits to being present during pitch sessions and the showcase of the games. This feedback and the greater availability of experts proved to be essential for the participants, as their suggestions and improvements regarding the integration of the

theme and its integration in playability also created more awareness among the participants for the relevance of the facilitators: *"I think that if it hadn't been for the guide, we would have been much more out of touch with the subject because in the eyes of those who don't know or who don't live here, who aren't used to it or who don't have that kind of knowledge or someone to share it with, the Lagoon is just a body of water"*. Furthermore, the cultural visit to Óbidos Lagoon was held almost three weeks before the event's start, which allowed for a greater spacing of days and, consequently, more time to reflect on the theme and research. Compared to **PT1**, where the cultural visit happened on the first day, and the participants also started the game-making process on that day, the participants from **PT2** were now more organized, and the ideation session went smoothly. In this CGJ, where both participants from Lusófona University and high school students were involved, it was difficult to foster relationships between the participants due to differences in age, background, studies and confidence. There was a clear division between them, which was only less felt during the showcase of the games on the last day. For the future, fostering this collaboration and willingness to connect between the students is important. Besides the communication challenges, the gender balance in **PT2** remained an issue - although there was an improvement - that needs to be further addressed and challenged in the next CGJs. One of the drawbacks of **PT2** was that the high school students had computers with insufficient capacity to run programs like Unity, which led to delays. Despite efforts to organize a Unity workshop to ensure all students could develop their games similarly and to create equal opportunities between high school and university students, the disparity in computer performance ultimately made a significant difference between the two groups.

4 Discussion

Toolkits provide generative, open-ended authoring within a design space, where its users can reuse, combine and adapt the building blocks provided by the toolkit. Ledo et al. summarize previous discussions ([15, 25, 30]) on the value of HCI toolkits into five goals: *Reducing Authoring Time and Complexity*, *Creating Paths of Least Resistance*, *Empowering New Audiences*, *Integrating with Current Practices and Infrastructures*, and *Enabling Replication and Creative Exploration*. A strength of the multiple-site case study presented in this paper is that it shares not only the same CGJ toolkit [29], collaborative and co-creative framing [27], and practical protocol [17] but also research protocol [37]. Evaluation methods that only examine a small subset of a toolkit or the results from its use may not demonstrate the research contribution, nor do they necessarily determine a toolkit's success [20]. However, with the broad data we have collected through the shared research protocol from the six cases, we are starting to see the many under-explored aspects of the large design space that the CGJ toolkit is trying to address and the many possibilities in this differentiated replication. Although the facilitation procedure, local themes, and situated aspects varied in the six cases, we saw several effects of using shared tools. First, the CGJs allow for the replication of ideas that explore a concept and, together, create a powerful pool of value-based proofs-of-concept that youth, as empowered new audiences, have authored in a cultural heritage context. We see how the CGJ process and toolkit

made it easier for youth to design games within a short amount of time by encapsulating concepts to simplify expertise. The youth were able to integrate the CGJ toolkit into their current practices, and through tools such as the expert council, they were able to define pathways to build on and extend solutions, leading them to the right solutions and away from the wrong ones.

Overall, the CGJ format worked according to its main objectives in the six CGJs. Or as one participant in DK1 states: *"It's really just a cool experience to come up with and make a fun game based on struggles or good values in our society"*. However, the research project is both an asset and a hindrance in the practical unfolding of the CGJs. The strength of the research project is its focus on knowledge creation, foundations in both theory and practice, a shared toolkit and research protocol, and systematic documentation and testing. With the multiple-case study, this paper indicates how its findings are relevant and applicable in diverse contexts and towards scaling it for reimagining and transforming youth experiences with cultural heritage, displaying straightforward, practical applications for research and development. Although we have some practical input for further format iteration, we also see some challenges. Striking a balance between the research project's and participants' needs is delicate. Certain things, such as paperwork and information concerning informed consent according to GDPR rules, etc., must be part of such a research project, although they tire the participants. If we were 'just' running it as a game jam without the research, then some of this paperwork would not be necessary. Although both in DK2 and NL2, the facilitation, paperwork and number of activities were reduced and streamlined based on experiences from the previous CGJs, there is still more to do to this. The CGJ toolkit needs further simplification and elaboration to truly support the participants' empowered participation in cultural heritage. However, an earlier research paper emphasised that participating youth experienced individual and collective cultural empowerment through playful interactions with(in) cultural heritage in CGJs [16]. Though iterations are still needed regarding the CGJ toolkit, its application thus displays signs of enabling youths engagement towards cultural and societal issues. This includes the procedure for documentation of the games. Some participants were proud of having their games published online in a global library⁸: *"I liked that all of the games were put up and made public on the website; it meant that I can now easily find and show others what we made together."* (youth in DK1), while others saw it as a necessary evil that took away time from the actual game development: *"The devlogs were certainly not necessary for such a short game and just hindered the process of actually making it. Itch.io was helpful to publish the game, though."* (youth in DK1):

Another challenge we met is having two distinct groups in the same CGJ, as in DK2 with high school students as part of an educational initiative and the volunteer jammers, or in PT2 with high school students and university students. This demonstrates that the CGJ toolkit is not equipped with support for mixing groups with various levels of competencies or pushing non-digital students to create hybrid or partly digital parts of their game concepts. This also surfaces the practical problem that the different groups have a variation in equipment (as in PT2), so perhaps the CGJ toolkit

should include computers or ensure that Unity or some other game engine is downloaded before the CGJ to provide equal accessibility.

Overall, the CGJ format and toolkit seem successful in supporting differentiated replication, with the quadruple helix cultural innovation system truly collaborating, especially after the necessary establishment of a local youth advisory board. Its participants were recruited from the initial CGJs, and the plan for the next round of six CGJs is that the local youth advisory boards will hopefully be mixed and provide more global input and perspectives to the whole helix.

5 Conclusion

In this case study paper, we used the Theoretical Framework for Game Jams in Applied Contexts to analyse a multiple-site case study of six cultural game jams in three countries. By focusing on the problem space, jam design, jam delivery and outcomes, and follow-on opportunities, we have been able to explore the practicalities of packaging, carrying out, and making sense of the first six locally arranged cultural game jams, which hopefully can feed into the design of cultural game jams of more global character. Since reporting differentiated replication in game jams with youth is rare, we contribute a unique illustration of variations in conditions and contexts and how these have influenced the planning, execution and evaluation of the related cases. In future work, an analysis of games should be conducted to evaluate how values and cultural heritage have been integrated by youth in games, and how the interplay of tangible and intangible natural and cultural heritage develops.

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⁸EPIC-WE game library <https://epic-we.au.dk/games>

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